TRADEWINDS RESUBMISSION DOSSIER // DOCKET 10-26-3976 (RESUBMISSION OF 9-26-3703)

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DOD CDAO TRADEWINDS SOLUTIONS MARKETPLACE // RESUBMISSION DOSSIER & VIDEO SCRIPT
ACTIVE DOCKET: 10-26-3976 (RESUBMISSION OF 9-26-3703) | READINESS: TRL 4 (TARGET: TRL 6)
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PART 1: 4-PILLAR RESOLUTION OF PRIOR DOCKET 9-26-3703 ASSESSOR FEEDBACK

[CRITERION-1] CRITIQUE : Defining the Problem: Quantify exact DoD infrastructure exposure, federal threat citations, and physical scan-latency failure.
RESOLUTION: Cited U.S. GAO reports GAO-21-81 & GAO-24-106137 (500+ DoD installations, 300,000+ buildings, $4.2B+ OT energy assets on 50-200ms PLCs) and Joint CISA/NSA/DoD Advisories AA23-144A & AA24-038A (Volt Typhoon OT pre-positioning) plus TRISIS/Triton. Proved physics gap: <4.16ms arc-flash backfeeds and 6.0ms microreactor excursions outrun 50-200ms PLCs by 8x to 33x.

[CRITERION-2] CRITIQUE : Accelerating the Mission: Clarify TRL 4 software testbed vs. TRL 6/7 hardware target, zero cloud dependency, and empirical test metrics.
RESOLUTION: Explicitly baselined at TRL 4 (Laboratory Software Emulation Testbed running on local workstation loopback 127.0.0.1 with 0% cloud dependency), targeting TRL 6 hardware demonstration in our Phase-1 OTA. Verified live in software across 636 C2 modules (1,961 Private / 1,770 Public), 85/85 passing unit test suites, 17/17 blocked Red-Team penetration vectors, 52 SQLite WORM & Memory tables, and 1,420 verified Field Academy engineering exam questions.

[CRITERION-3] CRITIQUE : Advancing the State of the Art: Provide a quantitative head-to-head Analysis of Alternatives (AoA) against commercial OT/SCADA & AI incumbents.
RESOLUTION: Contrasted Project Ebony’s TRL 4 zero-cloud software testbed and modeled 36.75ns Analog Kinetic Guillotine MOSFET design target (1,360,544x faster than a 50ms PLC, modeled at 28.5W on 24V DC) directly against Siemens S7-1500/Rockwell ControlLogix (50-200ms, 150-450W), Schneider Triconex SIS (10-50ms, 200-600W), passive IDS Dragos/Nozomi Guardian (0 physical cutoff authority, 300-1,200W), and cloud AI wrappers like Jev (>250ms RTT, fails under air-gap).

[CRITERION-4] CRITIQUE : Business Model & Transition: Present transparent TRL 4 -> TRL 6 commercialization roadmap, unbundled per-node CLIN pricing, and 10 U.S.C. 4022 OTA.
RESOLUTION: Replaced any ambiguous field-deployment language with transparent, audit-proof commercialization readiness: engineered an isolated client evaluation package (HVF-NODE-TEELE-001), staged draft Oklahoma Title 61 / HB 2992 & ODOT bridge drone proposal blueprints inside our repository, established our unbundled 26-Module Console CLIN Schedule (CLIN 0001 Base Pilot at $2,450/mo; CLIN 1004-A 1-Node Bundle at $895K/yr), and structured a 12-Month, $1.485M Phase-1 Prototype OTA under 10 U.S.C.

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4022 ($150K M1A Mobilization) to fabricate physical PCB hardware and transition to 10 U.S.C. 4022(f) production.

PART 2: QUANTITATIVE ANALYSIS OF ALTERNATIVES (COMPETITOR COMPARISON MATRIX)

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* METRIC: Primary Architecture
- Project Ebony (HVF 1AHA8) : Tri-Brain Software Testbed (TRL 4) + Modeled 36.75ns Analog Guillotine
- Siemens / Rockwell PLCs : Digital OS/RTOS PLC (5-20Hz Cyclic Scan)
- Dragos / Nozomi Guardian : Passive Network IDS / Cloud-Hybrid SIEM
- Schneider Triconex SIS : Triple-Modular Redundant Digital PLC
- Cloud AI Wrappers (Jev) : Cloud API Classification Wrapper
* METRIC: Cutoff Latency (Design Target vs Legacy)
- Project Ebony (HVF 1AHA8) : 36.75 ns (Modeled SPICE Target) / 2.04 us (Emulated Modbus FC05)
- Siemens / Rockwell PLCs : 50,000,000 - 200,000,000 ns (50 - 200 ms)
- Dragos / Nozomi Guardian : NONE (0 Physical Cutoff Authority)
- Schneider Triconex SIS : 10,000,000 - 50,000,000 ns (10 - 50 ms)
- Cloud AI Wrappers (Jev) : 250,000,000+ ns (> 250 ms Cloud API RTT)
* METRIC: Speed & Physics Advantage
- Project Ebony (HVF 1AHA8) : Modeled 1,360,544x Faster than 50ms PLC (Targets <4.16ms Arc-Flash)
- Siemens / Rockwell PLCs : 8x - 33x Too Slow for <4.16ms Arc-Flash / 6ms Excursion
- Dragos / Nozomi Guardian : Requires Human Operator (> 250 ms Reaction)
- Schneider Triconex SIS : Vulnerable to TRISIS/Triton Firmware Injection
- Cloud AI Wrappers (Jev) : Total Failure Under EMCON / SATCOM Severance
* METRIC: Target SWaP Power Draw
- Project Ebony (HVF 1AHA8) : 28.5 W Modeled Target (< 30W on 24V DC Tactical Bus)
- Siemens / Rockwell PLCs : 150 W - 450 W (Rack + HMI + Active Cooling)
- Dragos / Nozomi Guardian : 300 W - 1,200 W (Appliance Server + Cloud)
- Schneider Triconex SIS : 200 W - 600 W (TMR Chassis)
- Cloud AI Wrappers (Jev) : Data Center GPU Cluster Required
* METRIC: Cloud Dependency (Verified Live)
- Project Ebony (HVF 1AHA8) : 0% (100% Air-Gapped Localhost 127.0.0.1 Software Testbed)
- Siemens / Rockwell PLCs : Proprietary Firmware / Windows Engineering Workstation
- Dragos / Nozomi Guardian : High (Cloud Threat Feeds / Linux OS)
- Schneider Triconex SIS : Proprietary Firmware (Compromised in TRISIS)
- Cloud AI Wrappers (Jev) : 100% Dependent on External Commercial Cloud
* METRIC: WORM Cryptographic Ledger (Verified Live)

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- Project Ebony (HVF 1AHA8) : SHA-256 WORM Merkle DAG (52 Verified SQLite Tables on Disk)
- Siemens / Rockwell PLCs : Unauthenticated Modbus/CIP by Default
- Dragos / Nozomi Guardian : Mutable Syslog / SIEM Export
- Schneider Triconex SIS : Standard Event Buffer (Non-Merkle)
- Cloud AI Wrappers (Jev) : Mutable Cloud Database Logs

PART 3: SECTION VII.e COMPLIANT VIDEO TELEPROMPTER SCRIPT (5 SEGMENTS)

>>> SLIDE 1 (0:00 - 0:50) // RESUBMISSION GATE: 4-PILLAR ASSESSOR FEEDBACK RESOLUTION
RUBRIC: Mandatory Resubmission Compliance Gate (Docket 10-26-3976 Resubmission of 9-26-3703)
SCRIPT: I am Jeffery Humphrey, Founder, President, and CEO of Humphrey Virtual Farms LLC, CAGE Code 1AHA8, UEI S1M4ENLHTDH5. This video is our official resubmission for Project Ebony under Tradewinds Docket 10-26-3976, directly resolving all four pillars of Assessor Feedback from our prior submission 9-26-3703 with 100 percent engineering transparency. First, we explicitly baseline Project Ebony at TRL 4--a laboratory software emulation testbed and mathematical proof of concept running on local workstation loopback 127.0.0.1 with zero cloud dependency, targeting TRL 6 physical hardware demonstration under our proposed 12-month, $1.485 million Phase-1 Prototype OTA. Second, we quantify the DoD OT threat landscape citing GAO-21-81, GAO-24-106137, and CISA Volt Typhoon advisories. Third, we present a head-to-head Analysis of Alternatives against Siemens, Rockwell, Schneider Triconex, Dragos, Nozomi, and cloud AI. Fourth, we present our unbundled 26-module CLIN price schedule and Phase-1 OTA transition plan.

>>> SLIDE 2 (0:50 - 1:40) // CRITERION 1: DEFINING THE DOD PROBLEM (GAO & CISA VOLT TYPHOON)
RUBRIC: Criterion 1: Defining the Problem (Quantified DoD Exposure & Physics Latency Gap)
SCRIPT: Criterion 1: Defining the Problem. Per U.S. GAO reports GAO-21-81 and GAO-24-106137, the Department of Defense operates over 500 military installations, 300,000 buildings, and $4.2 billion in annual installation energy and microgrid assets relying on legacy commercial SCADA PLCs scanning at 50 to 200 milliseconds. Per Joint CISA, NSA, FBI, and DoD Advisories AA23-144A and AA24-038A, state-sponsored Volt Typhoon actors actively pre-position across critical energy, water, and transportation OT corridors. In TRISIS and Industroyer2, attackers corrupted controller logic while spoofing operator HMIs. Because high-voltage arc-flash backfeeds propagate in under 4.16 milliseconds and microreactor excursions occur in 6.0 milliseconds, 50-to-200

millisecond commercial PLC scan loops are 8 to 33 times too slow to arrest kinetic equipment destruction.

>>> SLIDE 3 (1:40 - 2:40) // CRITERION 2: ACCELERATING THE MISSION (TRL 4 SOFTWARE TESTBED)
RUBRIC: Criterion 2: Accelerating the Mission (Verified Zero-Cloud Software Emulation & 85/85 Tests)
SCRIPT: Criterion 2: Accelerating the Mission. Project Ebony version 8.0.0 is verified live today at TRL 4 on air-gapped 127.0.0.1 loopback with zero percent commercial cloud dependency, zero external APIs, and zero WAN listening ports. In our software emulation testbed and SPICE mathematical models, Brain One models deterministic safety boundary checks, 2.04-microsecond RS-485 Modbus FC05 actuation, and our 36.75-nanosecond Analog Kinetic Guillotine hardware design target. Brain Two executes local offline diagnostic and anomaly algorithms with zero authority to bypass Brain One safety interlocks. Brain Three seals every telemetry frame and operator command into an append-only SHA-256 WORM Merkle DAG across 52 verified bare-metal SQLite tables on disk. Our local repository is empirically verified across 636 C2 modules, 85 of 85 passing automated test suites, 17 of 17 blocked Red-Team penetration vectors with automatic tamper zeroization, and 1,420 verified Field Academy engineering exam questions.

>>> SLIDE 4 (2:40 - 3:40) // CRITERION 3: ADVANCING STATE OF THE ART (COMPETITOR AoA **MATRIX)**
RUBRIC: Criterion 3: Quantitative Analysis of Alternatives vs. Legacy PLCs, IDS & Cloud AI
SCRIPT: Criterion 3: Advancing the State of the Art. On screen is our Quantitative Analysis of Alternatives comparing Project Ebony directly against incumbent competitors. Siemens S7-1500 and Rockwell ControlLogix PLCs require 50 to 200 milliseconds per digital scan and draw 150 to 450 watts. Schneider Electric Triconex safety systems require 10 to 50 milliseconds, draw 200 to 600 watts, and were breached by firmware injection in TRISIS. Passive intrusion detection systems like Dragos Platform and Nozomi Guardian draw 300 to 1,200 watts and have zero physical cutoff authority, requiring over 250 milliseconds for human intervention, just like cloud AI wrappers such as Jev that fail when air-gapped. By contrast, Project Ebony combines a verified zero-cloud WORM software enclave with a modeled 36.75-nanosecond Analog Kinetic Guillotine MOSFET design target--1,360,544 times faster than a 50-millisecond PLC--at a modeled 28.5 watts on a 24-volt DC bus.

>>> SLIDE 5 (3:40 - 4:35) // CRITERION 4: BUSINESS MODEL, 26-CLIN SCHEDULE & $1.485M OTA
RUBRIC: Criterion 4: Commercialization Readiness, 26-Module CLIN Pricing & 10 U.S.C. **4022 OTA**
SCRIPT: Criterion 4: Business Model and Transition Pathway. Every commercial and state artifact in our repository is transparently presented as staged software and

engineering readiness: we have packaged a standalone Red-Team-certified client evaluation build, staged architectural compliance blueprints for civil bridge and pavement diagnostics modeling a reduction from $4,600 snooper-truck bridge inspections to $250 per autonomous drone sortie, and completed our 110-control NIST 800-171 and 800-82 architectural mapping, with formal third-party C3PAO and DoD SPRS auditing scheduled inside Phase 1. Our unbundled 26-module C2 console scales from CLIN 0001 Base Pilot at $2,450 per month and CLIN 1004-A Full 1-Node Bundle at $895,000 per year to our proposed 12-month, $1.485 million Fixed-Price Prototype OTA under 10 U.S.C. Section 4022, beginning with a $150,000 mobilization tranche to fabricate physical Cortex-M7 and silicon-carbide MOSFET boards and advance from TRL 4 software emulation to TRL 6 operational demonstration, while retaining 100 percent DFARS 252.227-7018 Background IP. Thank you.